

Synthesis and characterization of waterborne polyurethane-based marine antifouling coating modified via glycerol laurate

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Funding information

National Natural Science Foundation of China, Grant/Award Number: 52101385; Chongqing Municipal Youth Science and Technology Talent Training Project, Grant/Award Number: 2023AY40018

Abstract

Marine biofouling is a major international challenge since it is responsible for several environmental problems and economic losses. Developing a novel antifouling coating with good water resistance, adhesion, and controlled degradation performance is crucial to solving the problem. Herein, a series of waterborne polyurethane emulsions were synthesized using the prepolymerization chain expansion and self-emulsification method, with varying contents of glycerol laurate (GL) as the post-chain extender. The findings demonstrated a notable enhancement in the controllable degradation performance, adhesion, marine antifouling properties, and water resistance with an increase in the concentration of GL. In particular, the film exhibited the most optimal overall properties when the introduction amount of GL was 2 wt%. The absorption rate of the artificial seawater (ASW) reached a state of equilibrium when GWPU2 was immersed in ASW for 60 days, with a final value of approximately 18.5%. The release rates of the GWPU2 antifouling coatings were observed to be the lowest at approximately 120 days, with a rate of approximately 37.59 $\mu\text{g}/\text{cm}^2/\text{day}$. The long-term antifouling capability of the GWPU resins was demonstrated in marine field tests, wherein they exhibited sustained antifouling activity for a period exceeding 9 months. In particular, GWPU2 exhibited noteworthy antifouling performance.

KEYWORDS

Biodegradable, coatings, polyurethane

1 | INTRODUCTION

Marine biofouling, defined as the undesirable accumulation of microorganisms, marine animals, and plants on submerged surfaces, represents a significant challenge for maritime industries and other marine activities.¹ The presence of marine biofouling can increase the roughness and drag resistance of ships, leading to a decline in their

performance and fuel efficiency.² The application of marine antifouling coatings has been proved to be the most cost-effective and convenient method for the prevention of marine biofouling.³ Although self-polishing acrylate antifouling coatings have been widely used for marine antifouling,^{4,5} the relatively low deformation ability of the coatings and the poor adhesion to epoxy undercoats are two major concerns that hamper their

application.⁶ The coatings are prone to cracking under large deformation due to their brittle feature. Moreover, a tie coat is often applied first before the antifouling coating to overcome the cohesion problem, which results in a considerable expenditure of time and material.⁷

A copolymer with superior mechanical properties and effective self-polishing performance could be regarded as a promising component for marine antifouling coatings. Polyurethane has attracted increasing interest in marine antifouling coatings recently owing to their excellent deformation ability and self-polishing features.⁷ Conventional polyurethane marine antifouling coatings are solvent-based, comprising volatile organic compounds that might cause significant harm to human health and the ecological environment. There is an urgent need for developing polyurethane coatings with good antifouling ability and environmental friendliness.^{8,9}

Waterborne polyurethane (WPU) has become increasingly prevalent across a diverse range of industrial sectors, including leather, fabric, rubber, and so on, due to its nontoxic, nonflammable, and nonpolluting nature.¹⁰ While hydrophily was the major concern that prevented the practical application of WPU for marine antifouling.^{11–13} Researchers have devoted considerable effort to improving the water repellency of WPU films. For example, Lei et al.¹⁴ examined a series of WPUs utilizing a range of amine chain extenders, including ethylenediamine (EDA), diethylenetriamine (DETA), and triethylenetetramine (TETA). The findings indicated that the WPU prepared using DETA as a chain extender exhibited water repellency. In a related study, Li et al.¹⁵ examined a series of WPUs utilizing 3-aminopropyl triethoxysilane as the modified post-chain extender. The findings indicated that the polyurethane (PU) films exhibited remarkable water resistance when the introduction amount was 3 wt%. However, excessive crosslinking of WPU films would impede the degradation performance of the material, which represented a significant limitation for its potential application in marine antifouling. The works above provided practical ways to enhance water repellency at the expense of degradation performance. Besides, most of the investigated hydrophobic constituents were derived from nonrenewable petroleum. Gradually replacing petrochemicals as a source of high-quality chemical feedstocks with green and sustainable energy is of paramount importance.^{16,17}

In this paper, we aim to develop an eco-friendly WPU film for marine antifouling by incorporating a plant-based constituent—glycerol laurate (GL).¹⁸ The water repellency of the film is derived from a hydrophobic side chain by grafting the GL onto the main chain of WPU

instead of excessive crosslinking so that the degradation performance would not be affected. A series of WPU materials containing GL were prepared. The mechanical behaviors and degradation properties of the WPU films and coatings were examined. The marine antifouling capacity of the WPU was assessed by marine field trials. These results would contribute to the application of WPU film in the marine antifouling field.

2 | MATERIALS AND METHODS

2.1 | Materials

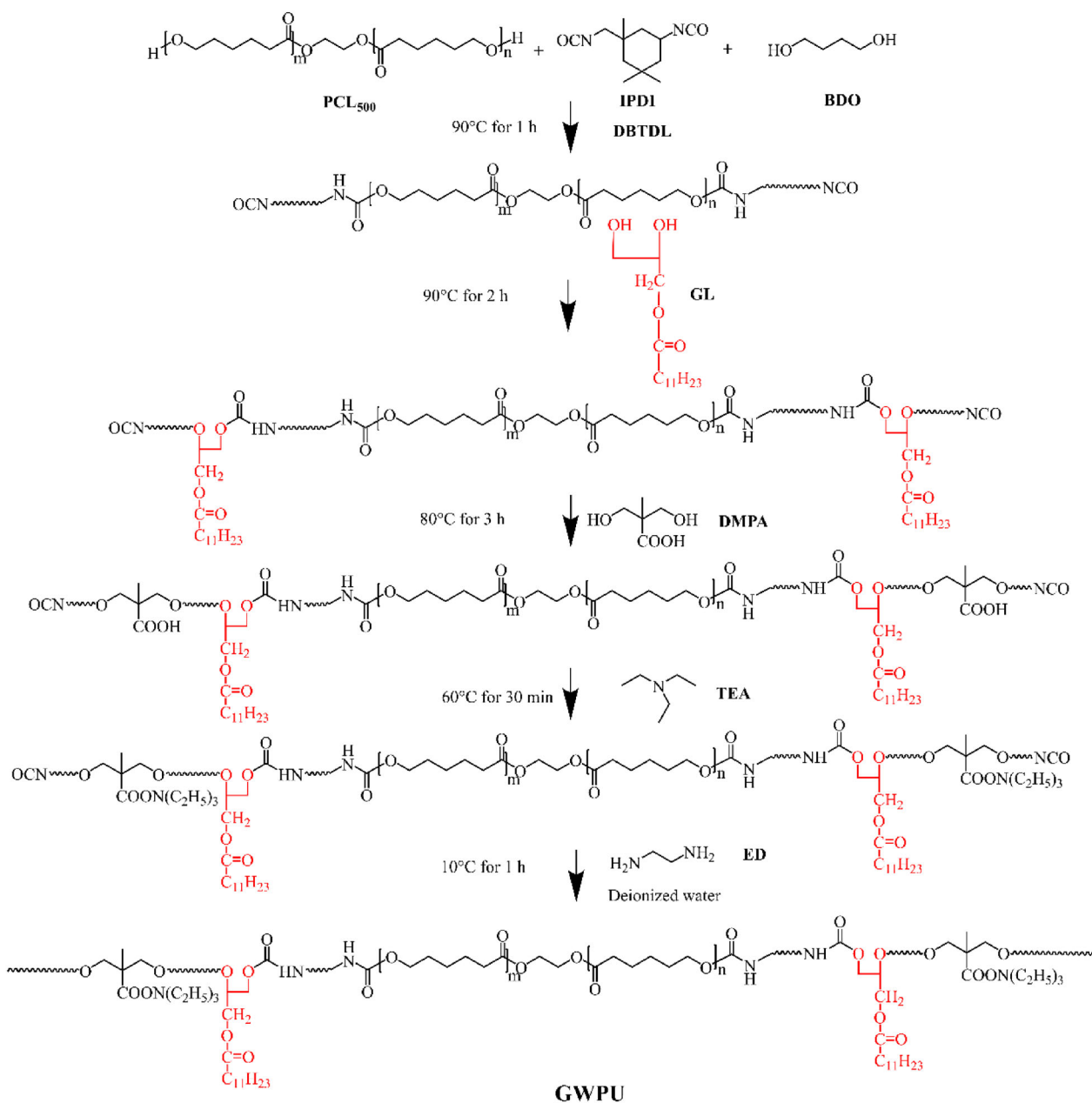
Polycaprolactone 500 (PCL500, $M_w = 500$ g/mol) was obtained from BASF(Germany) Co., Ltd. Isophorone diisocyanate (IPDI) was obtained from Covestro Polymers (China), LTD. Acetone, 2,2-bis (hydroxy-methyl) propionic acid (DMPA), glycerol laurate (GL), ethanediamine (ED), triethylamine (TEA), sodium hydroxide, dibutyltin dilaurate (DBTDL), 1,4-butanediol (BDO), and acetone were obtained from Aladdin (China) Biotechnology Co., Ltd. 4,5-Dichloro-2-octylisothiazolone (DCOIT) was obtained from Wuhan Jiacheng Chemical Co., LTD. UH-752 (thickening agent of waterborne polyurethane) was obtained from Shanghai Piyou New Material Co., LTD. Artificial seawater (ASW) was prepared by mixing sodium chloride (NaCl, 23 g/L), magnesium chloride hexahydrate ($MgCl_2 \cdot 6H_2O$, 9.8 g/L), sodium sulfate heptahydrate ($Na_2SO_4 \cdot 7H_2O$, 8.9 g/L), and calcium chloride dihydrate ($CaCl_2$, 1.2 g/L). All other reagents were of analytical grade and did not require further purification.

2.2 | Synthesis of GWPU

A series of GWPU samples with varying contents of GL were prepared and designated as GWPU0 through GWPU4, in accordance with the distinct GWPU percentages (0, 1, 2, 3, and 4 wt%) present in the total amount of starting materials. The compositions of each sample are presented in Table 1. Scheme 1 illustrates the route for synthesizing GWPU dispersions. Initially, IPDI, PCL500, BDO, and DBTDL were introduced into a three-necked flask equipped with a condensing apparatus and mechanical stirrer. The temperature should be maintained at 90°C for 1 h to obtain the prepolymer. Subsequently, GL should be added and allowed to react for 2 h. The temperature should then be adjusted to 80°C, and DMPA should be added and allowed to react for 3 h. Concurrently, acetone should be added to reduce the viscosity.

TABLE 1 Synthetic formulations of GWPU.

Sample	PCL500 (g)	IPDI (g)	BDO(g)	DMPA (g)	TEA (g)	GL (g)	ED (g)	Water (g)
GWPU0	40	46	1	5	3.7	0	2.3	233
GWPU1	40	46	1	5	3.7	1	2.3	233
GWPU2	40	46	1	5	3.7	2	2.3	233
GWPU3	40	46	1	5	3.7	3	2.3	233
GWPU4	40	46	1	5	3.7	4	2.3	233

SCHEME 1 Preparation route of GWPU dispersions. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

Subsequently, the temperature should be reduced to 60°C, and TEA should be introduced to neutralize the carboxylic acid groups. Finally, the GWPU dispersions

were created by emulsifying the mixture with deionized water and ED for 1 h at 10°C. The acetone was removed by rotary evaporation.

2.3 | Preparation of GWPU films

The GWPU emulsions were introduced into a Teflon mold and allowed to dry naturally for 2 days at 30°C. The resulting GWPU films were obtained by transferring them to vacuum drying at 100°C for a period of 24 h.

2.4 | Characterization of WPU

The stability of GWPU emulsions was evaluated via centrifugation (TG16A, China) at 3000 rpm for 30 min.

The size and zeta potential of GWPU particles were determined using a laser particle analyzer (NANO ZS90, Malvern, UK). The emulsions were diluted to approximately 0.01 wt% with deionized water at 25°C, and the laser scattering angle was set to 90°. Each sample was subjected to five independent tests.

The chemical structure of GWPU films was characterized using Fourier transform infrared spectroscopy with attenuated total reflection (FTIR ATR) (Vertex 70, Bruker, Germany). The frequency range was 4000–500 cm⁻¹, with 32 scans at a resolution of 4 cm⁻¹.

The thermal performance of GWPU films was identified using a differential scanning calorimeter analyzer (TA DSC25, USA). The GWPU films (5–7 mg) were heated from 20 to 200°C to eliminate their thermal history, then cooled to -30°C, and finally heated to 200°C at a rate of 10°C/min.

Wide-angle x-ray diffraction patterns were recorded on an x-ray diffraction instrument (XRD-7000, Shimadzu, Tokyo, Japan) using Cu K α radiation at 40 kV and 30 mA with a scan rate of 4°/min. The diffraction angle ranged from 5° to 40°.

The heat resistance of GWPU films was evaluated using a thermogravimetric analyzer (TGA550, USA) under an N₂ atmosphere. The temperature range for the test was set between 50 and 550°C, with a rate of 10°C/min.

The ASW contact angles of the GWPU films were determined via a dynamic contact angle analyzer (DSA30, KRÜSS, Germany) using the titration method. Each sample was subjected to five independent tests.

The tensile strength and elongation at the break of GWPU films were characterized at room temperature using a universal testing machine (AI-7000-SU1, China) with a strain rate of 100 mm/min. The GWPU films were sized at 50 mm × 5 mm × 0.5 mm, with the final result being the average of five parallel runs.

The GWPU films (50 mm × 50 mm × 0.3 mm) were immersed in an ASW solution at a temperature of 25°C. Subsequently, the samples were extracted from the ASW at predefined time intervals. Subsequently, the samples were weighed after being wiped and then again after

being dried in a vacuum at 60°C. For each data point, triplicate specimens were employed.

$$\text{The artificial seawater uptake (\%)} = \frac{w_2 - w_0}{w_0} \times 100, \quad (1)$$

$$\text{The weight loss (\%)} = \frac{w_1 - w_0}{w_1} \times 100, \quad (2)$$

where w_0 represents the dry weight of GWPU films after being vacuum dried, w_1 represents the initial weight of specimens, w_2 represents the wet weight of GWPU films after being wiped.

2.5 | Characterization of GWPU antifouling coatings

The GWPU antifouling coatings were premixed with DCOIT, which is a principal biocide in this formulation, and other ingredients as shown in Table 2 and Scheme 2, and processed in a paint conditioner for approximately 6 h.

The steel panels, coated with anticorrosion paints, were coated with GWPU antifouling coatings (with a thickness of approximately 300 μ m) without the use of additional tie paints. The pull-off adhesion data for the GWPU antifouling coatings was obtained by detaching an aluminum dolly at a speed of 0.2 MPa/s using a pull-off adhesion tester.

The method employed for the determination of the release rates of DCOIT from the GWPU antifouling coatings was based on the rotating cylinder method. The test cylinders (100 cm²) coated with GWPU antifouling coatings were immersed in ASW in a holding tank at 25°C. At specified time intervals, the test cylinders were exposed and rotated (60 rpm) at 25°C for 1 h in measuring containers (1500 mL of ASW). Subsequently, 100 mL of the measured ASW was extracted with 10 mL of hexane. The concentration of DCOIT was determined by UV-Vis spectroscopy (Agilent Cary 60) at a wavelength of 290 nm.

$$v = \frac{c \times 1500 \times 1 \times 24}{100}, \quad (3)$$

TABLE 2 The recipes of GWPU antifouling coatings.

Material	g/100 g coating
GWPU dispersions	47
DCOIT	39
Water	13
UH-752	1



SCHEME 2 Preparation route of GWPU marine antifouling coatings. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

where v is the release rate of DCOIT ($\mu\text{g}/\text{cm}^2/\text{d}$), c is the measured concentration of DCOIT.

The surface morphologies of GWPU antifouling coatings before and after being immersed for 120 days were examined using ULTRA55 scanning electron microscopy (SEM, GEMINI 500, Carl Zeiss Ltd., Germany), respectively.

The surface topography and roughness of the GWPU2 antifouling coatings before and after being immersed for 120 days were investigated by atomic force microscope (AFM, NX 10, PARK, Korea), respectively.

A series of marine field tests were conducted on PVC panels coated with GWPU antifouling coatings between September 2023 and May 2024. The tests were conducted at the inner Xiamen Bay ($24^{\circ}45'\text{N}$, $118^{\circ}07'\text{E}$) in China, a location with a high prevalence of marine biofouling due to the presence of a diverse array of fouling organisms.

3 | RESULTS AND DISCUSSION

3.1 | Emulsion performance of GWPU dispersions

Table 3 presents the data on the particle size distribution and emulsion properties of GWPU dispersions. It can be observed that the particle size of the GWPU dispersion exhibited an increase from 81.16 to 89.59 nm with the addition of GL, from 0 to 4 wt%. It was observed that the introduction of GL as a chain extender into the GWPU system increased the particle size of the GWPU emulsions. The GWPU emulsions exhibited a zeta potential value exceeding 40 mV, indicative of high storage stability. Furthermore, no precipitation was observed following centrifugation at 3000 rpm for 30 min, thereby confirming the excellent stability of the GWPU emulsions.

TABLE 3 The performance of the GWPU dispersions.

Sample	Z-average size (nm)	PDI	Zeta potential (mV)	Storage stability (month)
GWPU0	81.16	0.327	-43.26	>6
GWPU1	83.29	0.296	-46.85	>6
GWPU2	85.17	0.317	-47.49	>6
GWPU3	88.26	0.276	-48.24	>6
GWPU4	89.59	0.293	-46.92	>6

3.2 | The ATR-FTIR spectra of GWPU films

The ATR-FTIR spectra of GWPU films are shown in Figure 1. The characteristic band of the —NH— stretching vibrations was observed at 3327 cm^{-1} . The absorption bands at 2934 and 2848 cm^{-1} were attributed to the asymmetric and symmetric stretching of the methylene group ($\text{—CH}_2\text{—}$). The stretching absorption of the carbonyl group (urethane group) is observed at 1707 cm^{-1} , while the carbonyl group (PCL500) is observed at 1659 cm^{-1} . The characteristic band of the —CO—NH— group was observed at 1534 cm^{-1} . The characteristic band of the —C—O—C— group was observed at 1223 cm^{-1} .

3.3 | DSC analysis of GWPU films

Figure 2 presents a graphical representation of the thermodynamic properties of the GWPU films. The glass transition (T_g) temperatures of the GWPU films exhibited a gradual increase ($10.68\text{--}27.62^{\circ}\text{C}$) with the incorporation of greater amounts of GL (0–4 wt%). This was primarily because the rise in GL content resulted in a correlation

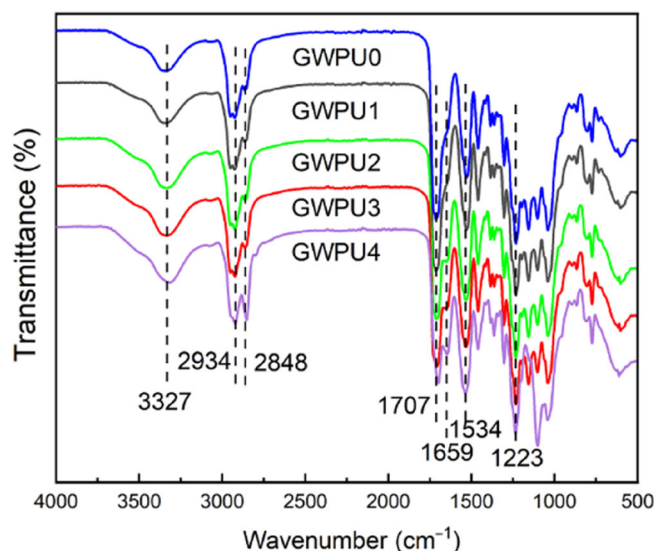


FIGURE 1 The attenuated total reflection-Fourier transform infrared spectra of the GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

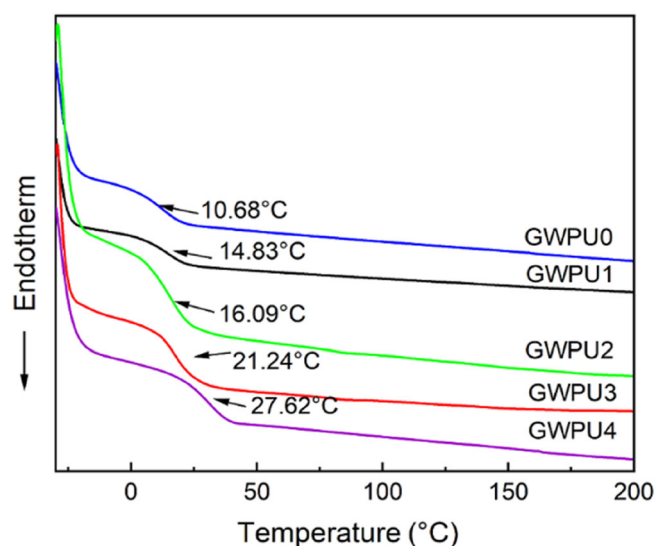


FIGURE 2 The differential scanning calorimeter plots of GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

with the hard segment, which was equivalent to an increase in the quantity of the hard segment. The quantity of urea bonding groups formed in the GWPU films augmented, thereby leading to an increase in the intermolecular and hydrogen bonding forces between the urea and urethane bonding groups.

3.4 | XRD analysis of GWPU films

To further verify the crystallization properties of the GWPU films, the XRD technique was employed. Figure 3

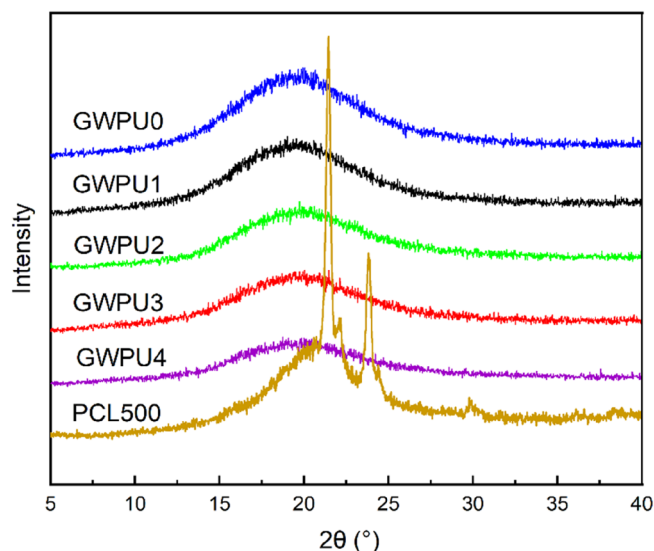


FIGURE 3 X-ray diffraction patterns of the GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

exhibits XRD curves of GWPU films and PCL500, respectively. The sharp peaks observed at 21.3° and 23.5° are indicative of the characteristic diffraction patterns exhibited by the PCL500 sample. The presence of these sharp peaks is indicative of the high degree of crystallinity exhibited by the PCL500 sample. In contrast, the GWPU films exhibit a broad diffuse peak with no discernible sharp diffraction peak, indicative of the dominance of amorphous structures over crystalline ones.¹⁹ This phenomenon can be attributed to the incorporation of the hard segment monomer (GL), which impedes the mobility of the molecular chain in GWPU films, thereby inhibiting their crystallization.

3.5 | TGA analysis of GWPU films

Figure 4 illustrates the thermogravimetric analysis (TGA) results for the GWPU films. All the GWPU films displayed three discrete stages of thermal degradation. The initial stage of degradation, occurring at approximately 200°C, was primarily attributed to the presence of free or hydrogen-bonded water molecules. The second stage (250–350°C) was primarily attributed to the degradation of urea and carbamate bonds. The third stage of thermal degradation (350–450°C) was attributed to the soft segments and residual small molecules present in the GWPU films. The findings of the study indicated that an increase in the GL content resulted in a decline in the thermal stability of GWPU films. This phenomenon can be attributed to the extended carbon chain of GL, which resulted in the formation of loosely structured GWPU films.

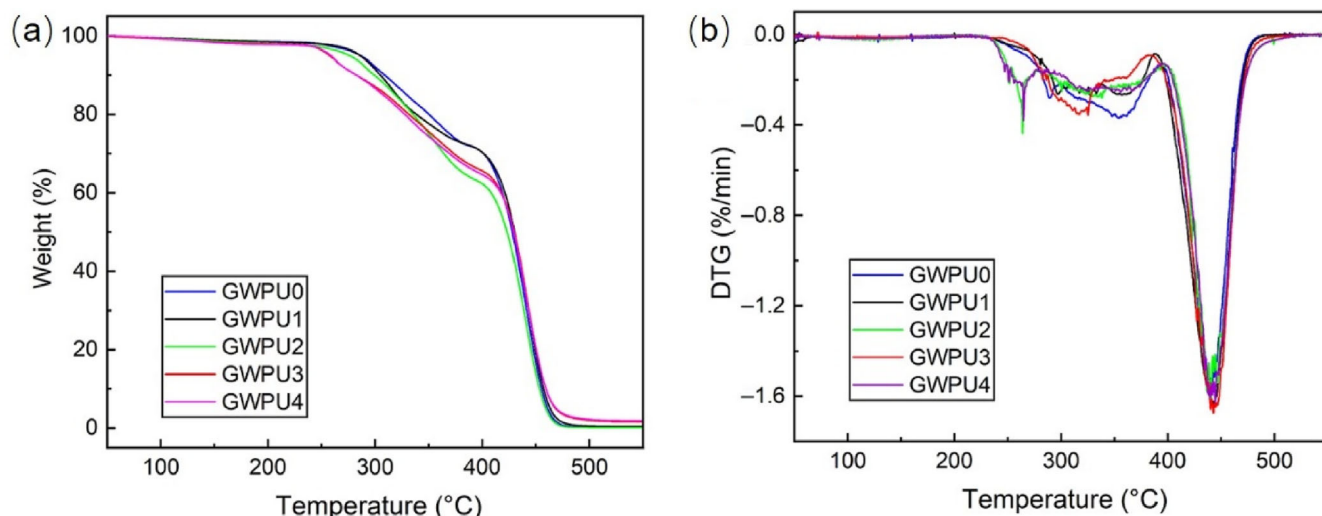


FIGURE 4 (a) Thermogravimetric analysis (b) Derivative thermogravimetry spectrum diagram of GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

TABLE 4 The mechanical properties of GWPU films.

Samples	Tensile strength (MPa)	Elongation-at-break (%)
GWPU0	7.26 ± 0.43	263 ± 22
GWPU1	7.11 ± 0.39	276 ± 19
GWPU2	6.86 ± 0.34	291 ± 21
GWPU3	6.64 ± 0.29	312 ± 19
GWPU4	6.49 ± 0.31	335 ± 22

3.6 | Mechanical properties of GWPU films

Table 4 illustrates the mechanical properties of GWPU films. It can be observed that the tensile strength of GWPU films decreased with an increase in the amount of GL. Consequently, the GWPU0 film exhibited the highest tensile strength of all the GWPU films. However, the elongation at the break of the GWPU films demonstrated an increase in the amount of GL. This phenomenon can be attributed to the extended carbon chain of GL, which resulted in the formation of loosely structured GWPU films.

3.7 | ASW contact angles of GWPU films

Figure 5 illustrates the ASW contact angle of the GWPU films. The GWPU4 film exhibited the largest contact angle, primarily due to its elevated GL content, which is comprised of long carbon side chains, rendering it hydrophobic. However, after being immersed in ASW

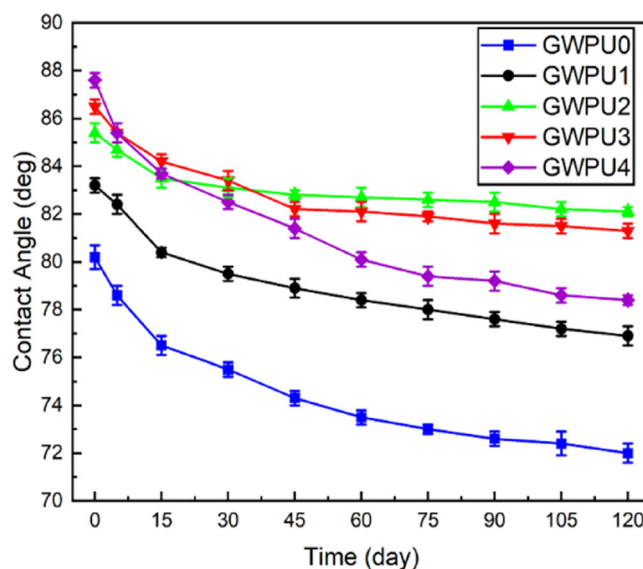


FIGURE 5 The artificial seawater contact angle of GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

for 60 days, the ASW contact angle of the GWPU4 film was observed to be smaller than that of the GWPU2 film. This can be attributed to the fact that the GWPU4 film exhibited the highest GL content, which resulted in a more porous film. As a result, the ASW was more readily able to penetrate the interior of the GWPU4 film, leading to degradation and the formation of a certain amount of carboxyl groups, which served to reduce its hydrophobicity. In contrast, the GWPU2 film exhibited a moderate amount of GL, which resulted in a more stable ASW contact angle with increasing immersion time.

3.8 | ASW absorption of GWPU films

Figure 6 illustrates the ASW absorption of GWPU films. It was observed that the ASW absorption of GWPU films for 5 days exhibited a decline with an increase in GL content. The presence of long carbon chain side chains in GL rendered it hydrophobic. However, after being immersed in ASW for 120 days, it was observed that the ASW absorption of GWPU4 was relatively higher than that of GWPU2. This phenomenon may be attributed to the fact that GWPU4 has the highest GL content, which resulted in a more porous GWPU4 film. Consequently, the ASW was more prone to penetrate the interior of the GWPU4 film. In contrast, when GWPU2 was immersed in ASW for 60 days, the rate of water absorption by the ASW reached equilibrium, reaching approximately 18.5%.

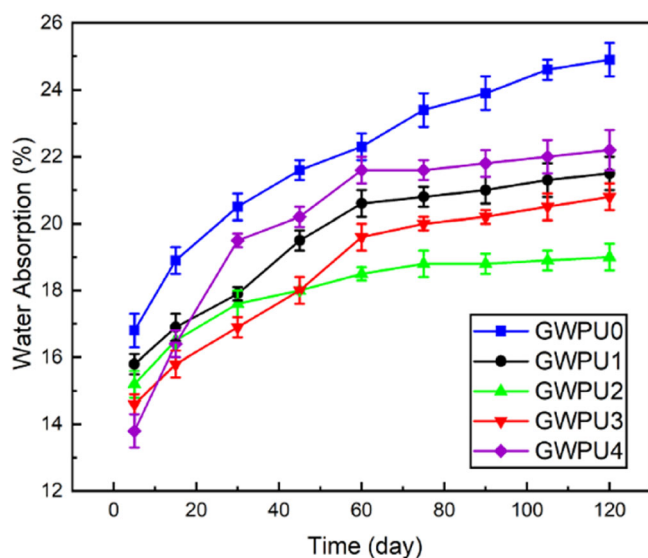


FIGURE 6 The artificial seawater absorption of GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

3.9 | Weight loss of GWPU films

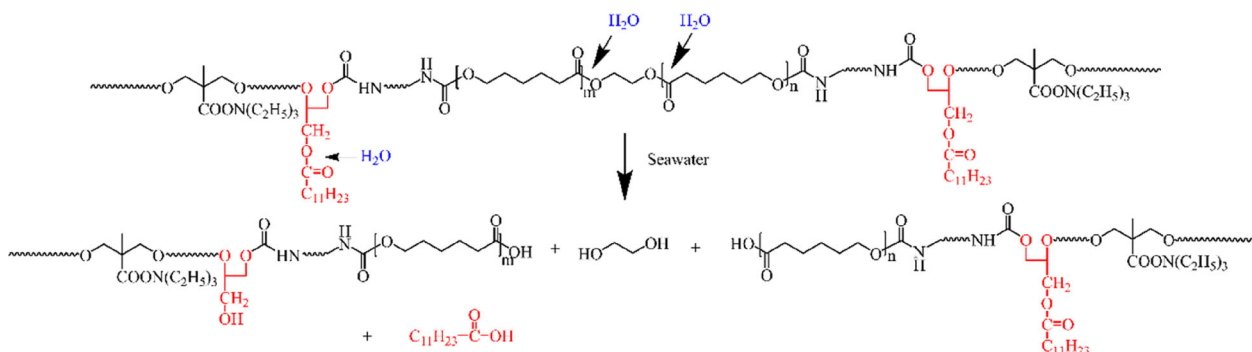
The weight loss of GWPU films was evaluated to assess their potential applications as marine antifouling coatings. The weight loss of GWPU films reflected the release of soluble oligomers from the GWPU films, as illustrated in Scheme 3. Figure 7 illustrates the weight loss of GWPU films in an ASW environment. It was observed that there was a notable correlation between the weight loss of GWPU films and the ASW absorption of GWPU films. The weight loss rate of GWPU films demonstrated a tendency towards stabilization.

3.10 | The adhesion strength of GWPU antifouling coatings

Figure 8 illustrates the adhesion strength of the GWPU antifouling coatings. It was observed that the initial adhesion strength of the GWPU antifouling coatings decreased with an increase in the amount of GL. After being immersed in ASW for 120 days, the adhesion strength of the GWPU antifouling coatings was observed to decrease. The GWPU2 antifouling coating exhibited the least reduction in adhesion strength (3.17 MPa), which may be attributed to the fact that this resin film exhibited the greatest resistance to degradation in ASW.

3.11 | The DCOIT release rates of GWPU antifouling coatings

The relationship between immersion time and DCOIT release rates of GWPU antifouling coatings is illustrated in Figure 9. It was observed that the DCOIT release rates of GWPU antifouling coatings exhibited a marked increase as the weight loss rate of GWPU films increased. The DCOIT release rates of GWPU antifouling coatings



SCHEME 3 The weight loss scheme of GWPU films. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

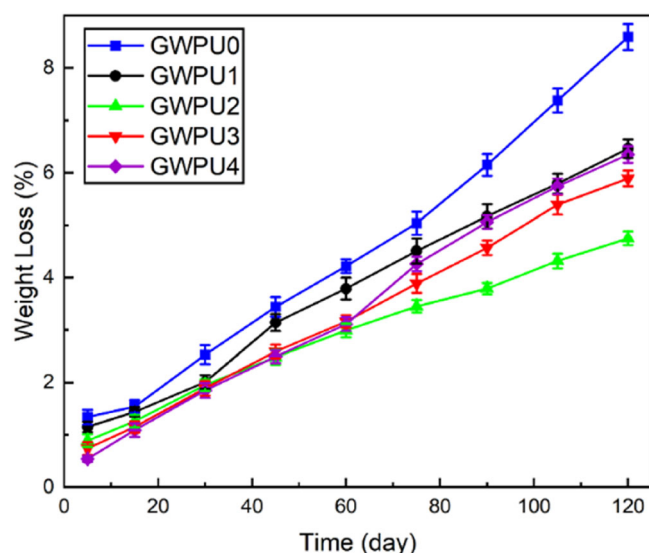


FIGURE 7 The weight loss of GWPU films in artificial seawater. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

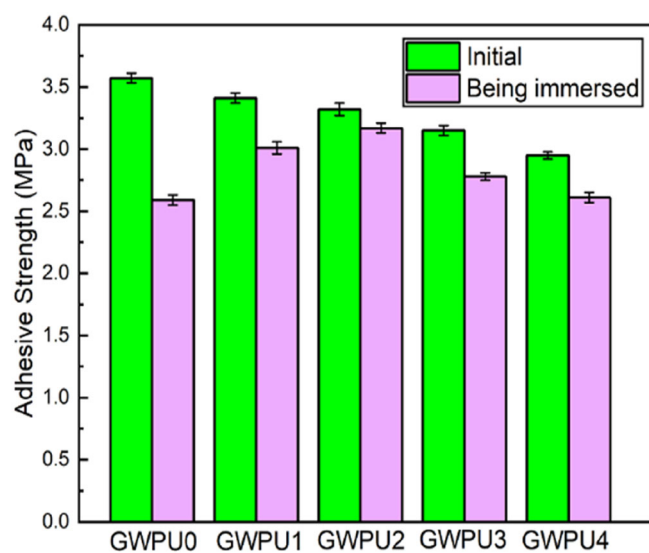


FIGURE 8 The adhesion strength of the GWPU antifouling coatings. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

remained constant for approximately 60 days. The DCOIT release rates of the GWPU2 antifouling coatings were observed to be the lowest at approximately 120 days (approximately $37.59 \mu\text{g}/\text{cm}^2/\text{day}$), which is conducive to marine antifouling.

3.12 | Surface analysis of GWPU antifouling coatings by SEM

Figure 10 depicts the scanning electron microscope images of the GWPU antifouling coating surface before

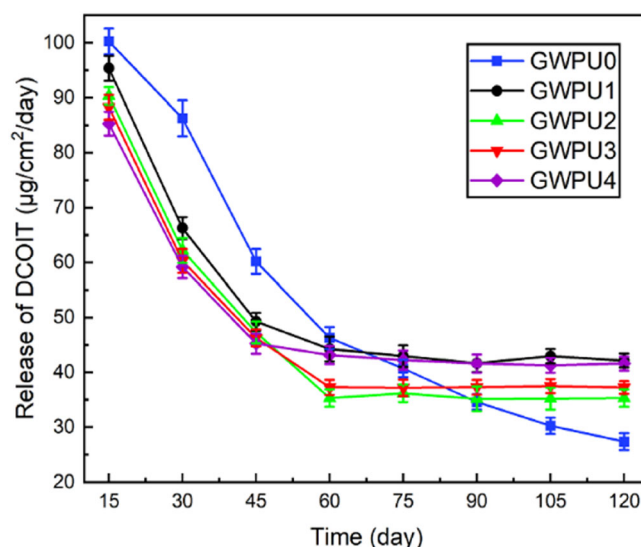


FIGURE 9 The DCOIT release rates of GWPU antifouling coatings. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

and after being immersed for 120 days, respectively. Before immersion in the ASW, the surfaces of the GWPU antifouling coatings displayed a relatively rough texture. It was observed that the GWPU antifouling coatings underwent hydrolysis of the ester linkages when exposed to ASW. Furthermore, the initial roughness of the GWPU antifouling coatings demonstrated a gradual reduction.

3.13 | Surface analysis of GWPU2 antifouling coatings measured by AFM

Table 5 illustrates the surface morphologies of the GWPU2 antifouling coatings before and after being immersed for 120 days, respectively. The average surface roughness was calculated to be 15.563 nm for GWPU2 antifouling coatings before being immersed, while the average surface roughness was calculated to be 2.122 nm for GWPU2 antifouling coatings after being immersed for 120 days. It was observed that the GWPU2 antifouling coatings became smoother due to the degradation of the GWPU2 film.

3.14 | Assessment of antifouling performance

The objective of this study was to examine the efficacy of GWPU antifouling coatings in preventing the attachment of marine organisms. After 9 months of immersion in seawater, the surfaces of the GWPU antifouling coatings (GWPU1, GWPU2, and GWPU3) were observed to be essentially free of contamination, as illustrated in

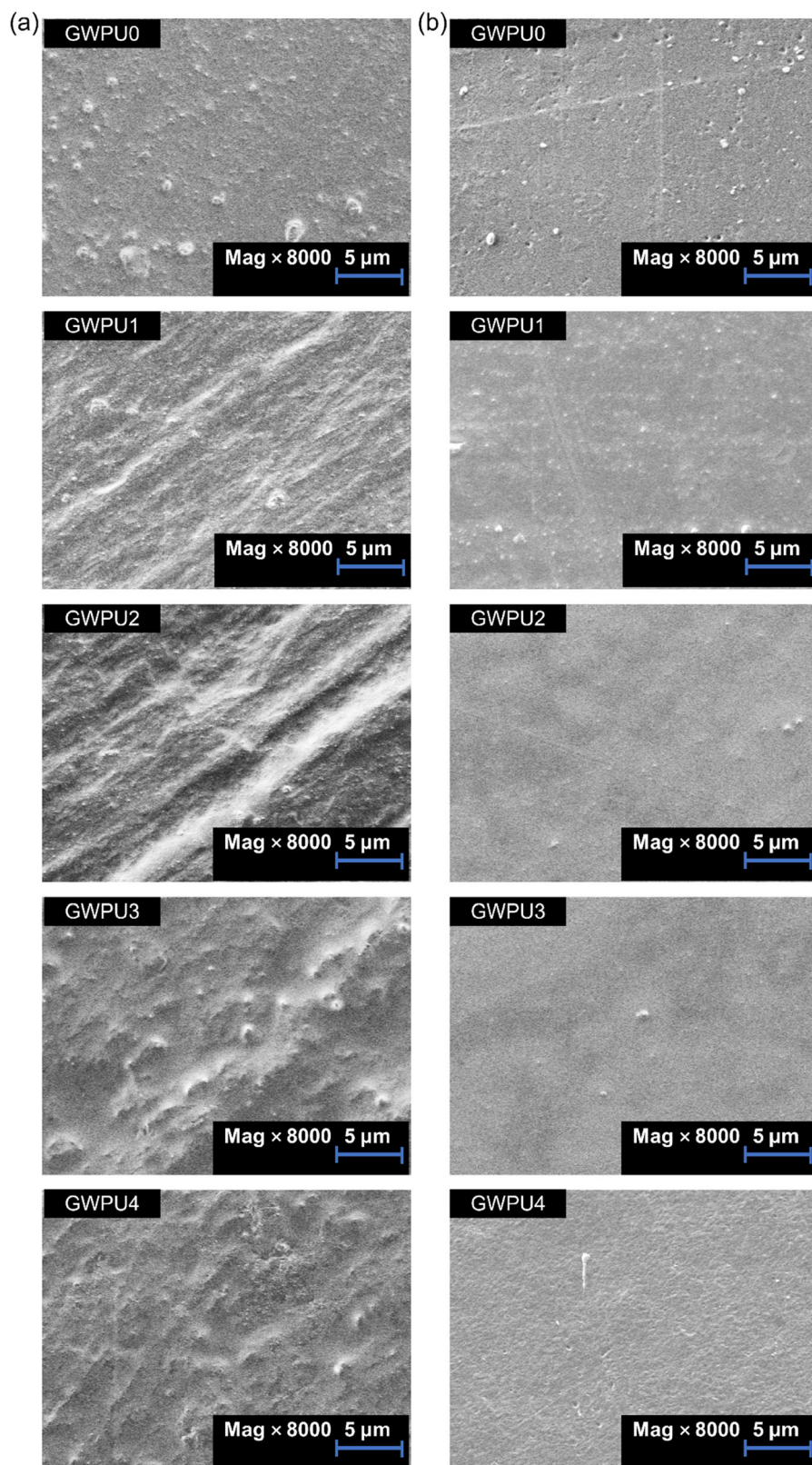
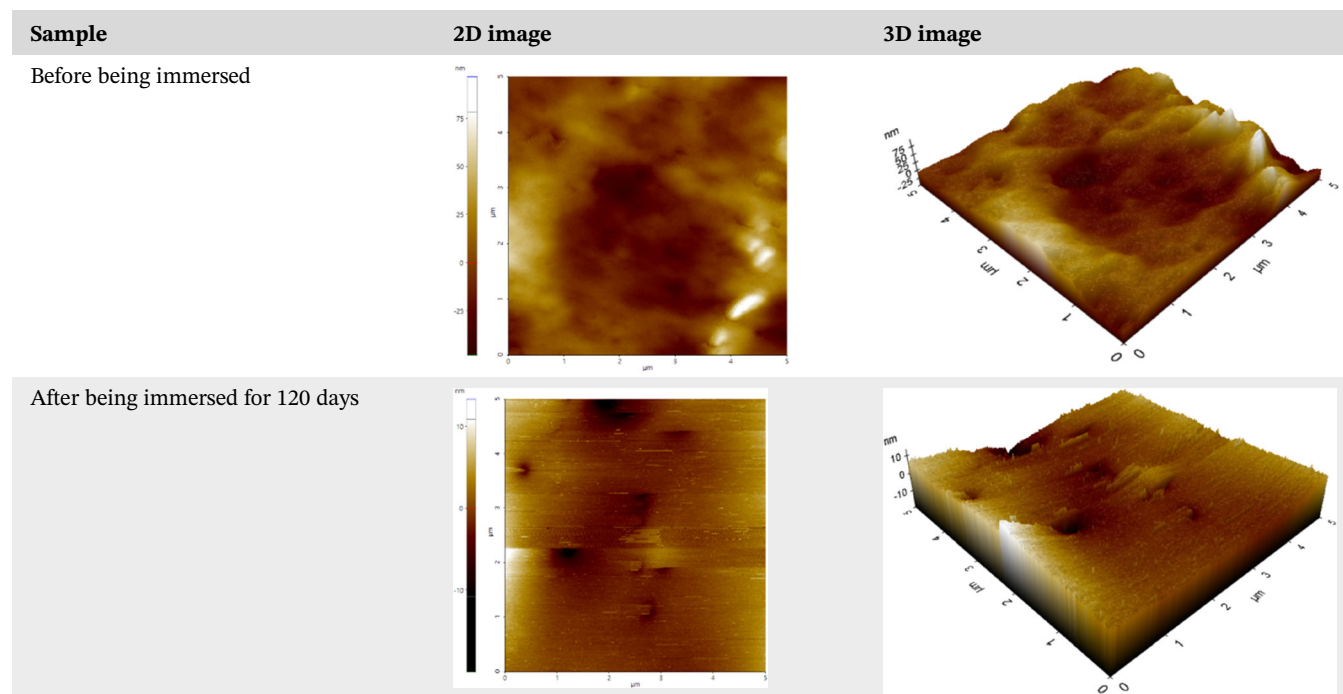


FIGURE 10 The scanning electron microscope image of GWPU antifouling coatings surface: (a) before being immersed, (b) after being immersed for 120 days. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.56455)]

Table 6. The results demonstrate that GWPU antifouling coatings (GWPU1, GWPU2, and GWPU3) exhibited the properties of an efficacious marine antifouling agent, as evidenced by the uniform and discernible concentration

of DCOIT observed at the coating surfaces. In consideration of these findings, it can be proposed that GWPU copolymers are an appropriate selection for incorporation into marine antifouling applications.

TABLE 5 The surface morphologies of the GWPU2 antifouling coatings measured by AFM. [Color table can be viewed at wileyonlinelibrary.com]



Abbreviation: AFM, atomic force microscope.

TABLE 6 The typical images of panels with GWPU antifouling coatings for marine antifouling. [Color table can be viewed at wileyonlinelibrary.com]

Time	GWPU0	GWPU1	GWPU2	GWPU3	GWPU4	Control sample
0 month						
9 month						

4 | CONCLUSION

A series of anionic WPU dispersions with varying GL contents were successfully synthesized via the internal emulsification method. The GL was used as a modified post-chain extender. The properties of anionic WPU dispersions, films, and marine antifouling coatings with various GL contents were systematically investigated. The particle size testing demonstrated that all dispersions exhibited particle sizes of less than 100 nm, indicating optimal storage stability. The FTIR-ATR spectra

indicated that the GWPU films were successfully prepared. Furthermore, the T_g values, the thermal stability, and the tensile strength of GWPU films were markedly diminished with the incorporation of GL. In particular, the film exhibited the most optimal overall properties when the introduction amount of GL was 2 wt%. The rate of absorption of the ASW reached a state of equilibrium when GWPU2 was immersed in ASW for 60 days, with a final value of approximately 18.5%. The GWPU2 antifouling coating exhibited an adhesion strength of approximately 3.17 MPa when immersed in ASW for 120 days.

The release rates of the GWPU2 antifouling coatings were observed to be the lowest at approximately 120 days, with a rate of approximately $37.59 \mu\text{g}/\text{cm}^2/\text{day}$. It was observed that the GWPU antifouling coatings underwent hydrolysis of the ester linkages when exposed to ASW. Furthermore, the initial roughness of the GWPU antifouling coatings demonstrated a gradual reduction. The long-term antifouling capability of the GWPU resins was demonstrated in marine field tests, wherein they exhibited sustained antifouling activity for a period exceeding 9 months. In particular, GWPU2 marine antifouling coating exhibited noteworthy antifouling performance.

In this paper, we have demonstrated the long-term antifouling capability of the GWPU film in a certain marine field (inner Xiamen Bay ($24^{\circ}45'\text{N}$, $118^{\circ}07'\text{E}$) in China). While the antifouling capability might be affected by different marine environments such as temperature, pH, salt content, and so on. In future work, we would like to consider these factors and compare the antifouling capability of the GWPU film in different serving environments.

AUTHOR CONTRIBUTIONS

Miao Cao: Data curation (lead); formal analysis (lead); funding acquisition (lead); writing – original draft (lead); writing – review and editing (lead). **Qiaoyi Xiao:** Writing – original draft (equal); writing – review and editing (equal). **Bo Wu:** Data curation (equal). **Zhisheng Luo:** Data curation (equal). **Zhengwei Dai:** Data curation (equal). **Jie Yi:** Data curation (equal); formal analysis (equal); funding acquisition (lead); writing – original draft (equal); writing – review and editing (equal).

ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (No. 52101385); the Jiaxing Youth Science and Technology Talent Project (No. 2023AY40018).

CONFLICT OF INTEREST STATEMENT

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY STATEMENT

Data will be made available on request.

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How to cite this article: M. Cao, Q. Xiao, B. Wu, Z. Luo, Z. Dai, J. Yi, *J. Appl. Polym. Sci.* **2025**, *142*(5), e56455. <https://doi.org/10.1002/app.56455>